

InternSpark — ML Classification Project

Task 1: Supervised Classification Model

1. Objective

Build and evaluate a supervised binary classification model, compare at least two algorithms, and report standard classification metrics.

2. Dataset

Breast Cancer Wisconsin Diagnostic dataset from scikit-learn: 569 observations, 30 numeric features and two target classes.

3. Methodology

80/20 stratified split; StandardScaler + Logistic Regression; Random Forest; 5-fold Stratified Cross-Validation; held-out test evaluation using Accuracy, Precision, Recall, F1 and ROC-AUC.

4. Test-set Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.9825	0.9861	0.9861	0.9861	0.9954
Random Forest	0.9474	0.9583	0.9583	0.9583	0.9937

5. Cross-Validation Summary

Logistic Regression

ACCURACY: mean=0.9780, std=0.0098

PRECISION: mean=0.9795, std=0.0162

RECALL: mean=0.9860, std=0.0131

F1: mean=0.9825, std=0.0078

ROC-AUC: mean=0.9959, std=0.0050

Random Forest

ACCURACY: mean=0.9626, std=0.0164

PRECISION: mean=0.9718, std=0.0086

RECALL: mean=0.9684, std=0.0233

F1: mean=0.9700, std=0.0135

ROC-AUC: mean=0.9923, std=0.0049

6. Visual Results

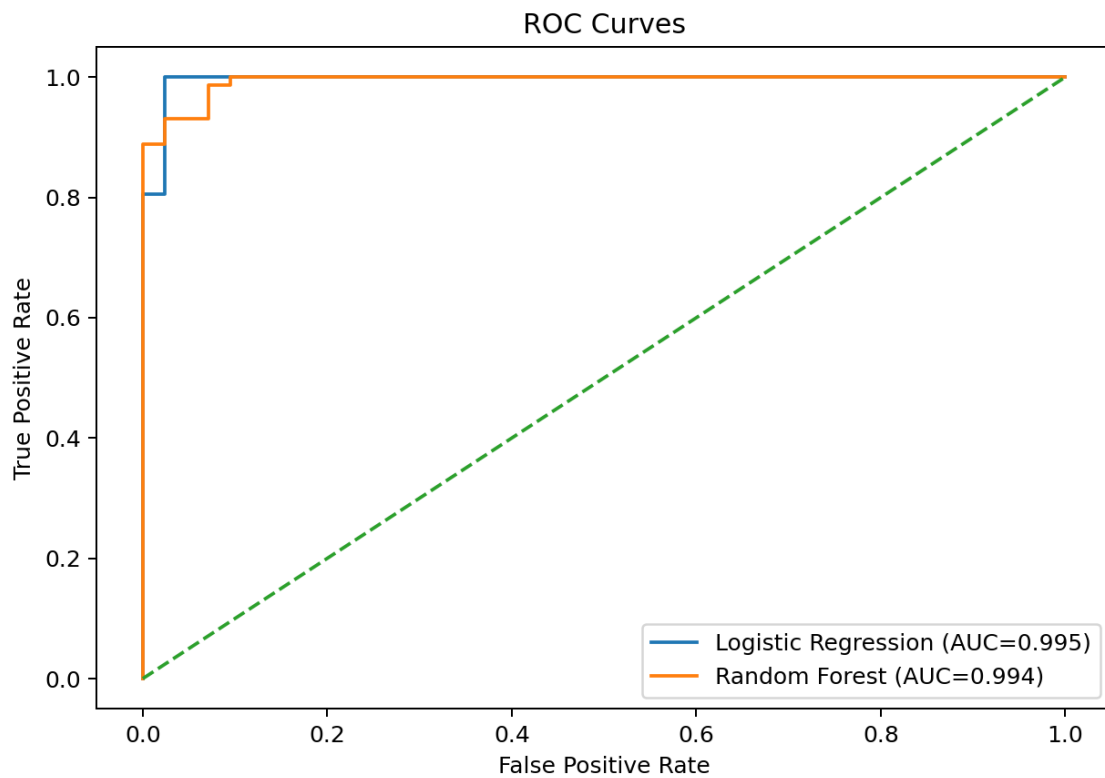


Figure 1: ROC curves.

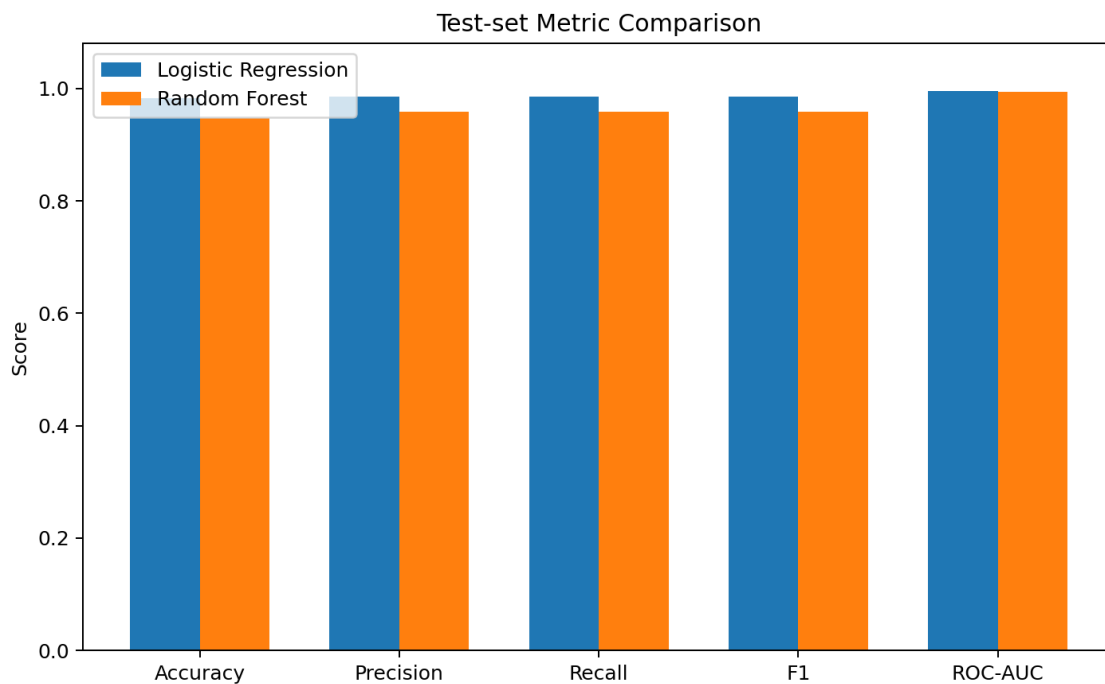


Figure 2: Test-set metric comparison.

7. Interpretation

Both classifiers perform strongly on the held-out test set. Precision, recall, F1 and ROC-AUC should be considered together rather than relying on accuracy alone.

8. Conclusion

The requested workflow is complete: preprocessing, train/test split, cross-validation, two-model comparison, required metrics and visual evaluation.

9. Reproducibility

Install requirements.txt and run ML_Classification_Project.ipynb or classification_project.py. Random seeds are fixed at 42.